Kunj Shah

kunjcr2@gmail.com linkedin.com/in/kunjcr2 github.com/kunjcr2 kunjcr2.github.io

Education

San Jose State University **May 2027**
Bachelor of Science in Computer Science *GPA: **3.88/4.00**, Dean's List*

Coursework: OOP, Advanced DSA, Computer Architecture, Information Security, ML, Software Engineering

Experience

A10 Networks **May 2026 – Aug 2026**
Deep Learning Engineer Intern *San Jose, CA*

- Architected and trained **TrooperViz** for multimodal safety evaluation across text, image, and multi-turn inputs, scaling PyTorch training across **8 GPUs** and **160,157 examples**.
- Fine-tuned a vision-language model and merged it with a text-only guardrail using **SLERP**; achieved **91.83% single-turn** and **90.24% multi-turn** accuracy on heldout multimodal benchmarks across **21 safety policies**, beating tested public guardrails by up to **16.4 percentage points**. Works similarly better on Multilingual benchmarks too.
- Built data and evaluation pipelines with **Pandas** and **Polars**, generating **85,000 synthetic conversations** from a **340,473-row data pool** and enabling interactive inference through **Gradio** and **vLLM**.

Routes Technologies **Oct 2025 – Present**
AI Engineer Intern *Remote*

- Architected an LLM-powered **NL-to-SQL** pipeline using few-shot prompting, input normalization, **470** synonym mappings, parameterized SQL sanitization, session memory, and **Weights & Biases** observability, deploying the system across **5 Azure ML endpoints**.
- Built an **SVD collaborative-filtering** recommendation system with scikit-learn and an LLM-based taste profiler, generating updated user preference profiles through automated nightly ETL jobs and achieving **86.7**% Precision@5 and **89.8**% Recall@10 on a **6**-user, **87**-recipe benchmark.
- Engineered a multi-source recipe ingestion pipeline using **Scrapy**, social-media APIs, LLM classification, language filtering, and **Faster-Whisper** transcription to process video and web-based content across **5** source integrations.

Research under Prof. Mark Stamp **Oct 2026 – Present**
Adversarial Neural Networks for Cryptography *San Jose State University*

- Investigating how neural-network architecture affects attacker performance in adversarial neural cryptography by comparing **4 architectures** across controlled cryptographic experiments spanning **5 attacker capacity levels (5K–80K parameters)** and **3 random seeds**.
- Implemented and evaluated attacker architectures in **PyTorch**, analyzing convergence, key-recovery behavior, and cross-architecture generalization across **57 of 60 completed experimental conditions** and **20,000 training updates**; observed MLP attacker accuracy increase from **55.73% to 63.83%** as capacity increased.

Dreamable Inc. **May 2025 – Aug 2025**
AI/ML Engineering Intern *San Francisco, CA*

- Fine-tuned a **Qwen 7B** model using TensorFlow, Hugging Face, and Lambda, applying **LoRA** to reduce training cost and memory usage while delivering a domain-specific question-answering model.
- Evaluated model configurations across multiple hyperparameters through systematic experimentation, tracking training and evaluation metrics with **Weights & Biases** to optimize model performance.
- Developed an AI outreach agent using **LangChain**, **Exa.ai**, and LLM APIs to automate messaging workflows, reducing manual outreach time and improving response rates while documenting processes in Notion.

Leadership / Extracurricular

AI Club at SFSU, *Vice President* Jun **2025** – Dec **2025**
SparkSF, *Tech Lead* Jun **2024** – Dec **2025**
Duc Ta, CS215, *Teaching Assistant* Jan **2024** – Dec **2024**
CodePath, *Advanced DSA Batch* Jan **2026** – May **2026**

Technical Skills

Programming: Python, SQL, Java, JavaScript, Git
Machine Learning & NLP: PyTorch, TensorFlow, scikit-learn, Weights & Biases, Pandas, NumPy, Neural Networks, Reinforcement Learning, LangChain
LLMs: Transformers, Pretraining, Fine-tuning, Inference, vLLM, Alignment, Benchmarking, Multi-GPU Training
Backend & Cloud: Flask, FastAPI, Node.js, PostgreSQL, MySQL, Docker, CI/CD, Microsoft Azure, Lambda